



Air Conditioning System Diagnosis

Quick Training Guide – QT711A



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Toyota Air Conditioning System Diagnosis

Quick Training Guide QT711A

Normal Operation

Low Charge

Overcharged

Compressor Failure

Moisture in System

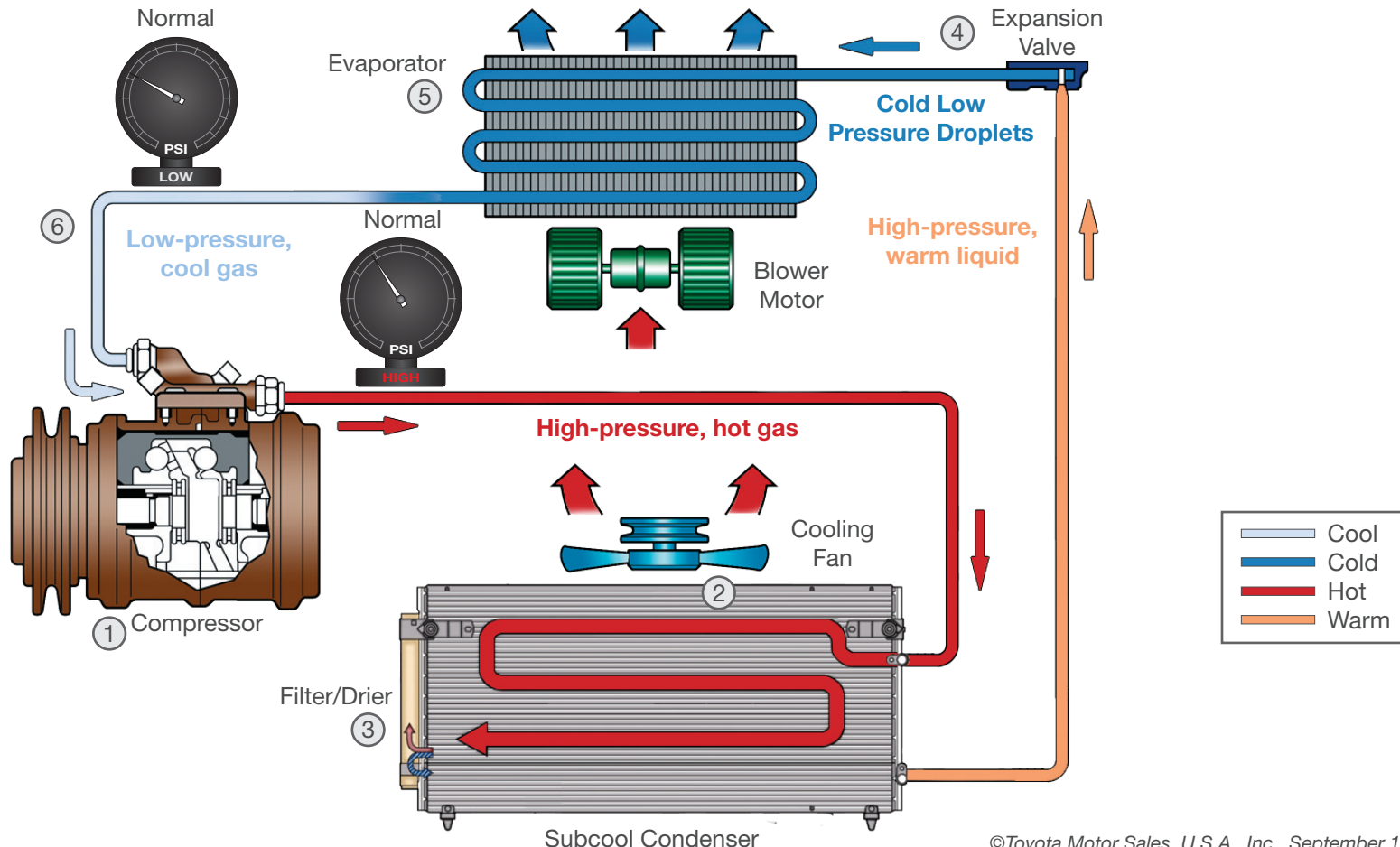
Restriction in
Condenser

Normal Operation

 Click the tabs at the top to view each section

1. The **compressor** pressurizes the refrigerant into a high pressure hot gas. The **condenser** removes heat from the refrigerant gas, turning it into a warm liquid.
2. The **cooling fan** provides air flow to assist in removing heat.
3. The **filter/drier** (or receiver-drier in some vehicles) filters impurities and water from liquid refrigerant.

4. A small opening in the **expansion valve** allows high-pressure liquid refrigerant to escape into the low-pressure side of the system. As it does, it expands, becoming a cool mist.
5. The **blower motor** blows air across the **evaporator** where heat from the air converts the refrigerant droplets into gas. This absorbs heat and cools the air.
6. The refrigerant and absorbed heat return to the compressor to repeat the cycle.



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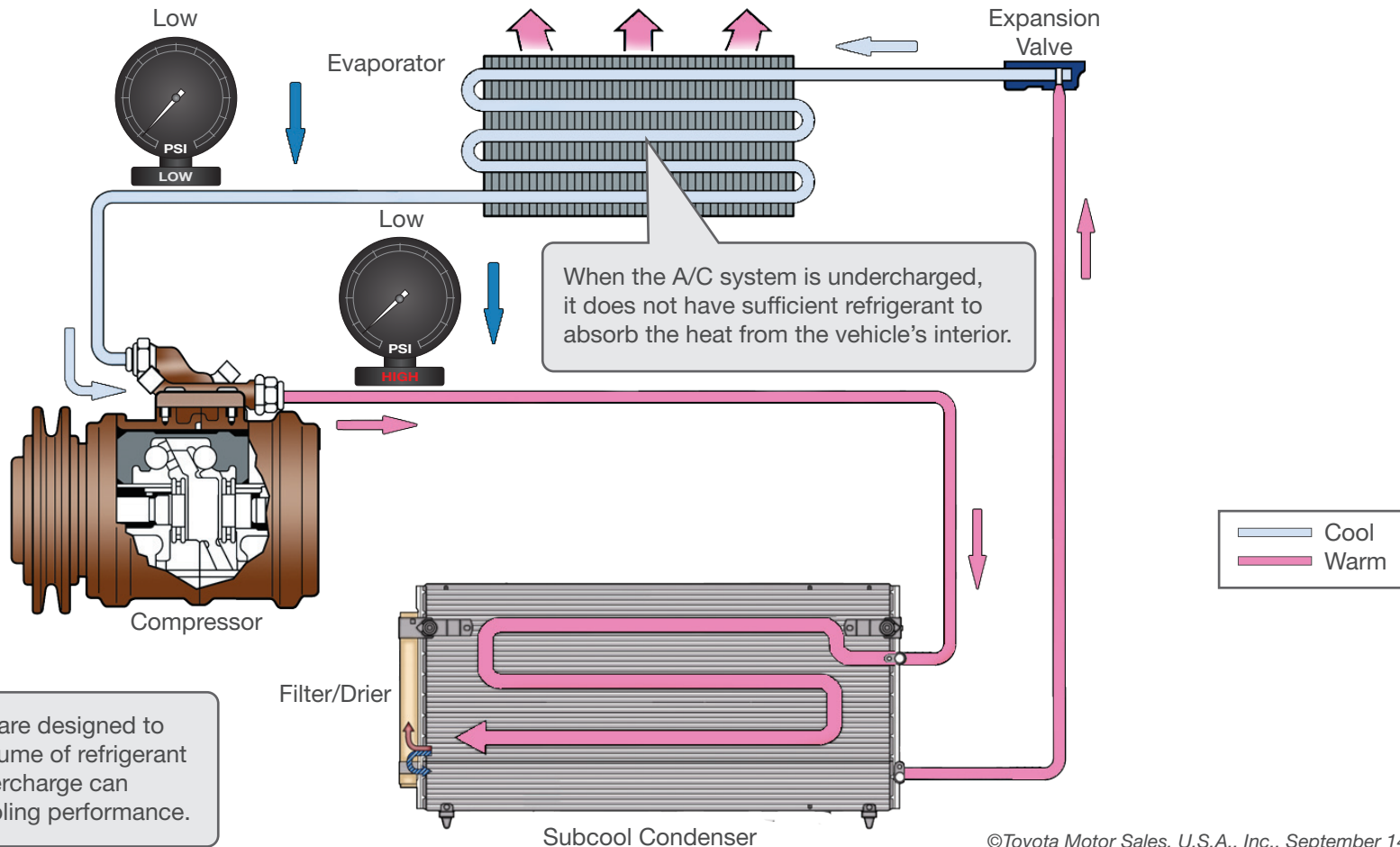
Moisture in System

Restriction in
Condenser

Low Charge

Symptoms:

- Poor cooling, no cooling at all
- Low side **below** normal
- High side **below** normal
- Air bubbles in sight glass (when equipped with a sight glass)



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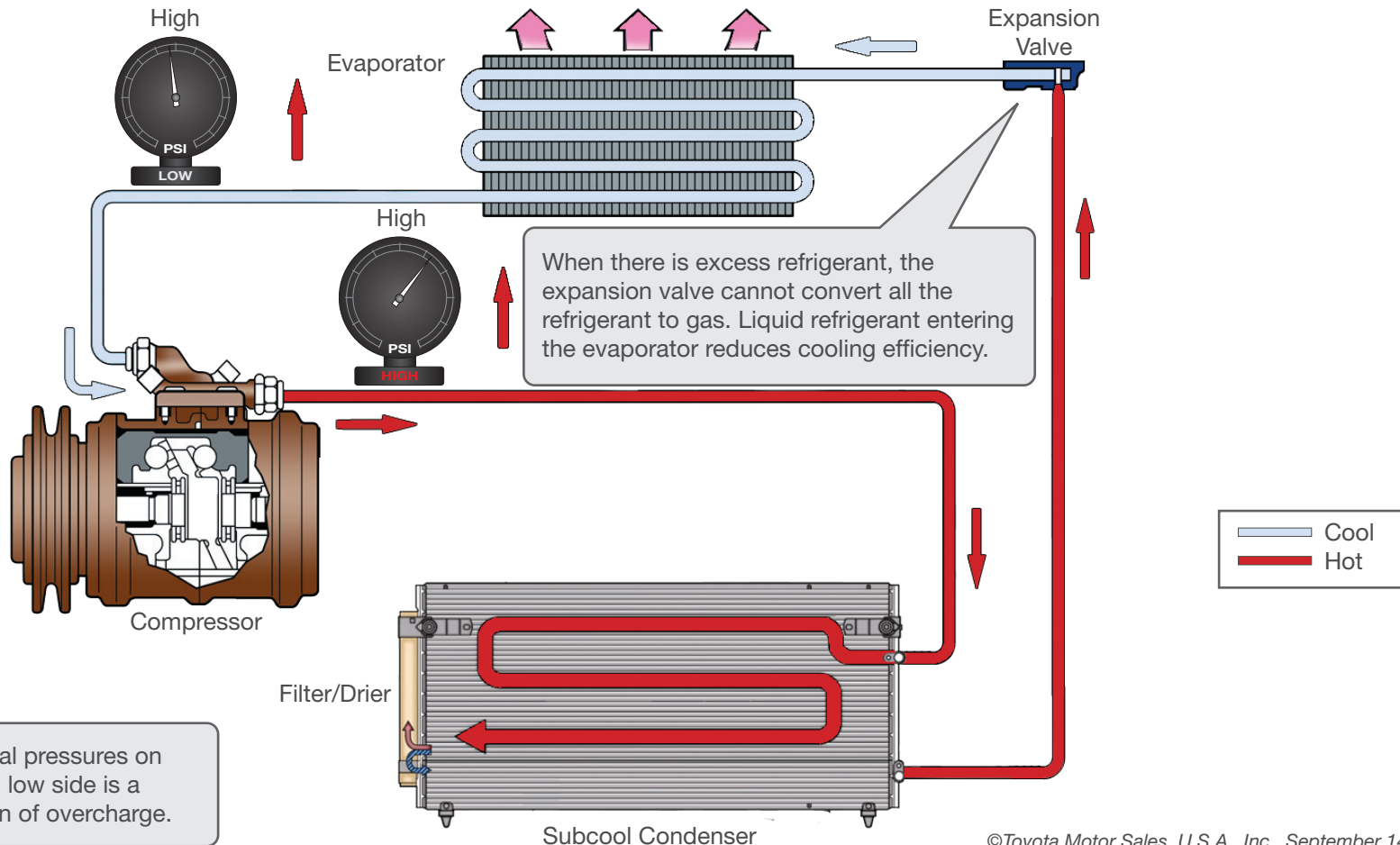
Moisture in System

Restriction in
Condenser

Overcharged

Symptoms:

- Poor cooling
- Low side **above** normal
- High side **above** normal
- No air bubbles in sight glass (when equipped with a sight glass)



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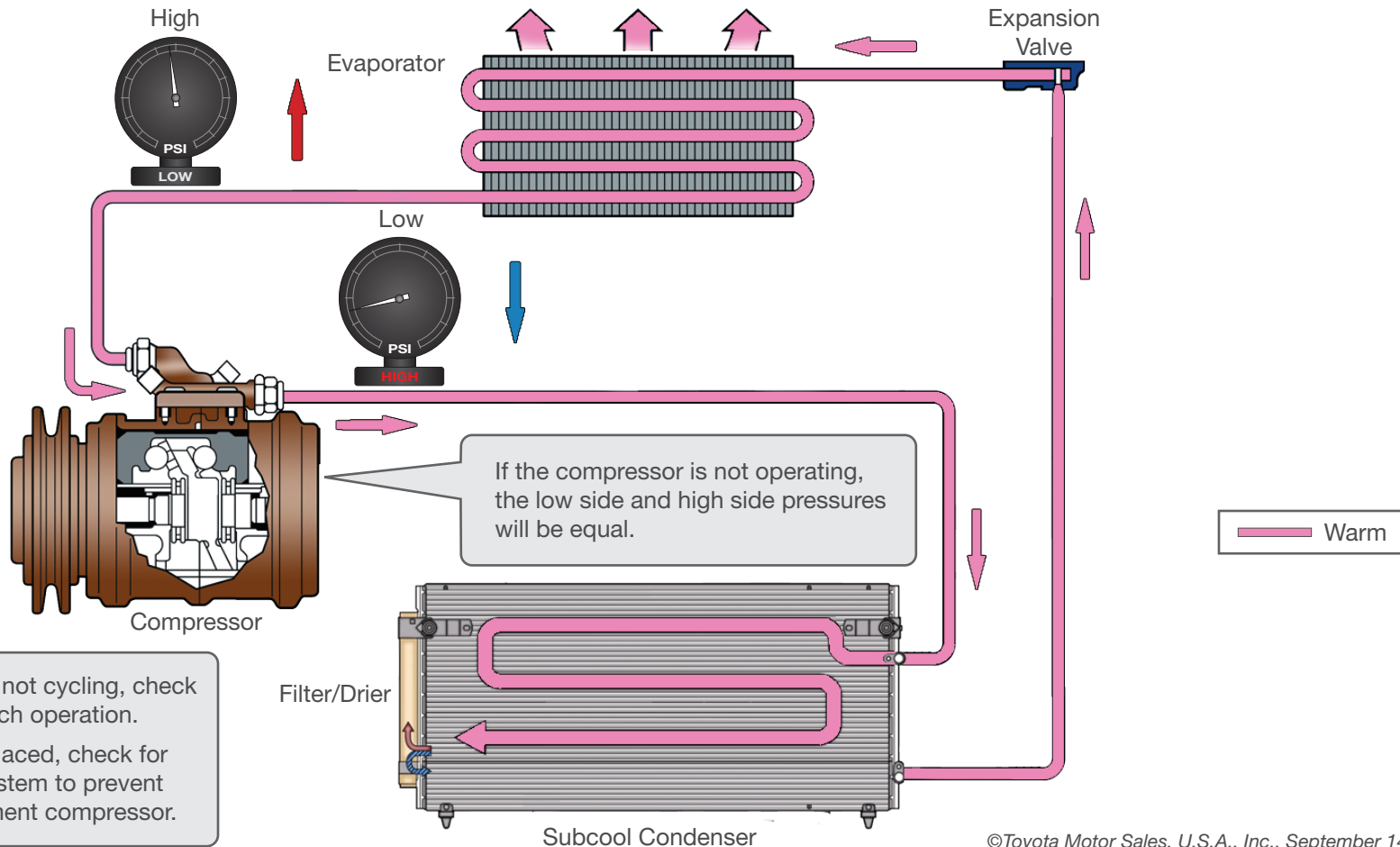
Moisture in System

Restriction in
Condenser

Compressor Failure

Symptoms:

- No cooling at all
- Low side **above** normal
- High side **below** normal
- No air bubbles in sight glass (when equipped with a sight glass)



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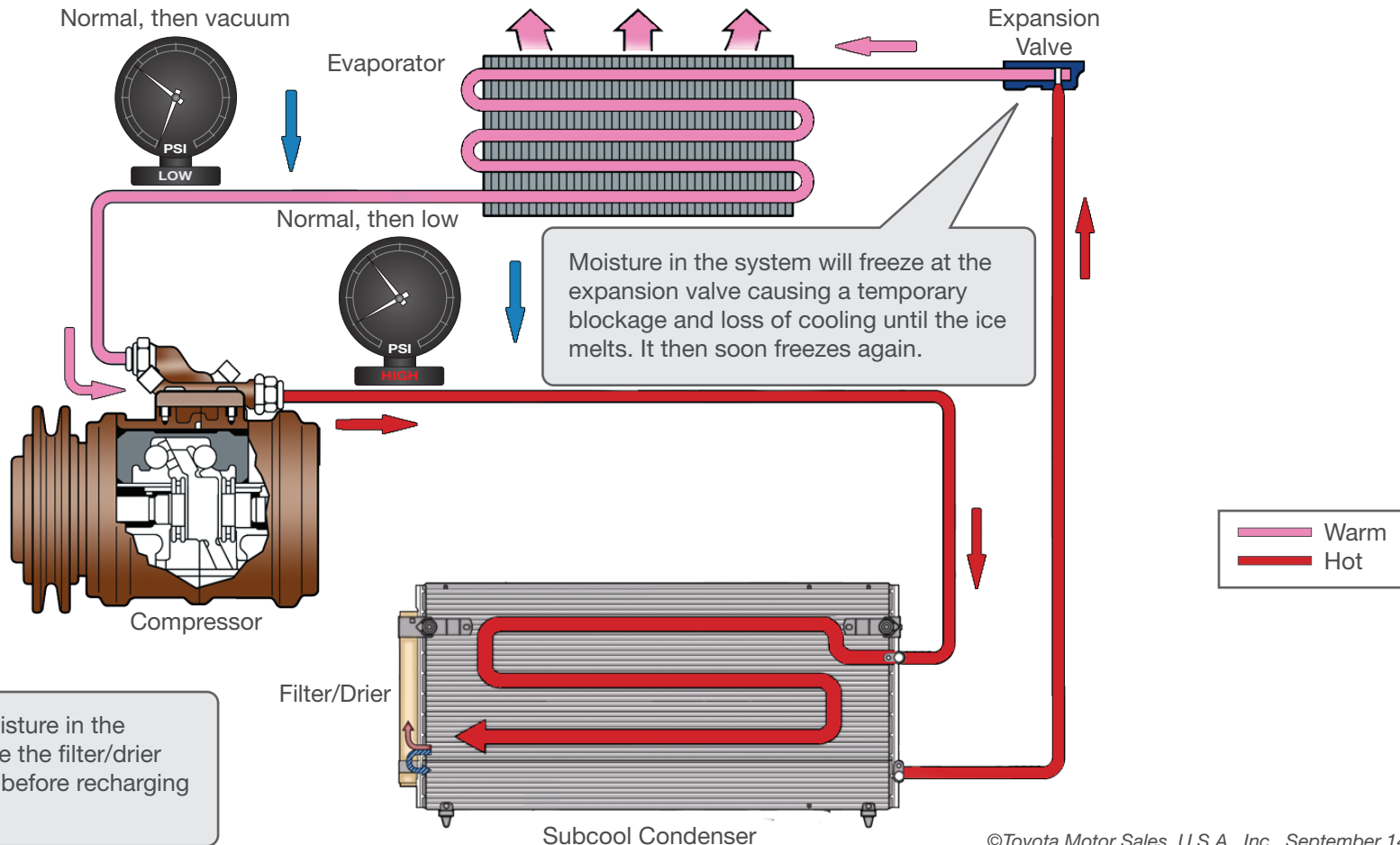
Moisture in System

Restriction in
Condenser

Moisture in System

Symptoms:

- Intermittent poor cooling
- Low side **at vacuum** intermittently
- High side **below** normal intermittently
- Pressures are normal for a while, then below normal for a while.
- Air is cool when pressures are normal, then not very cool when pressures drop.



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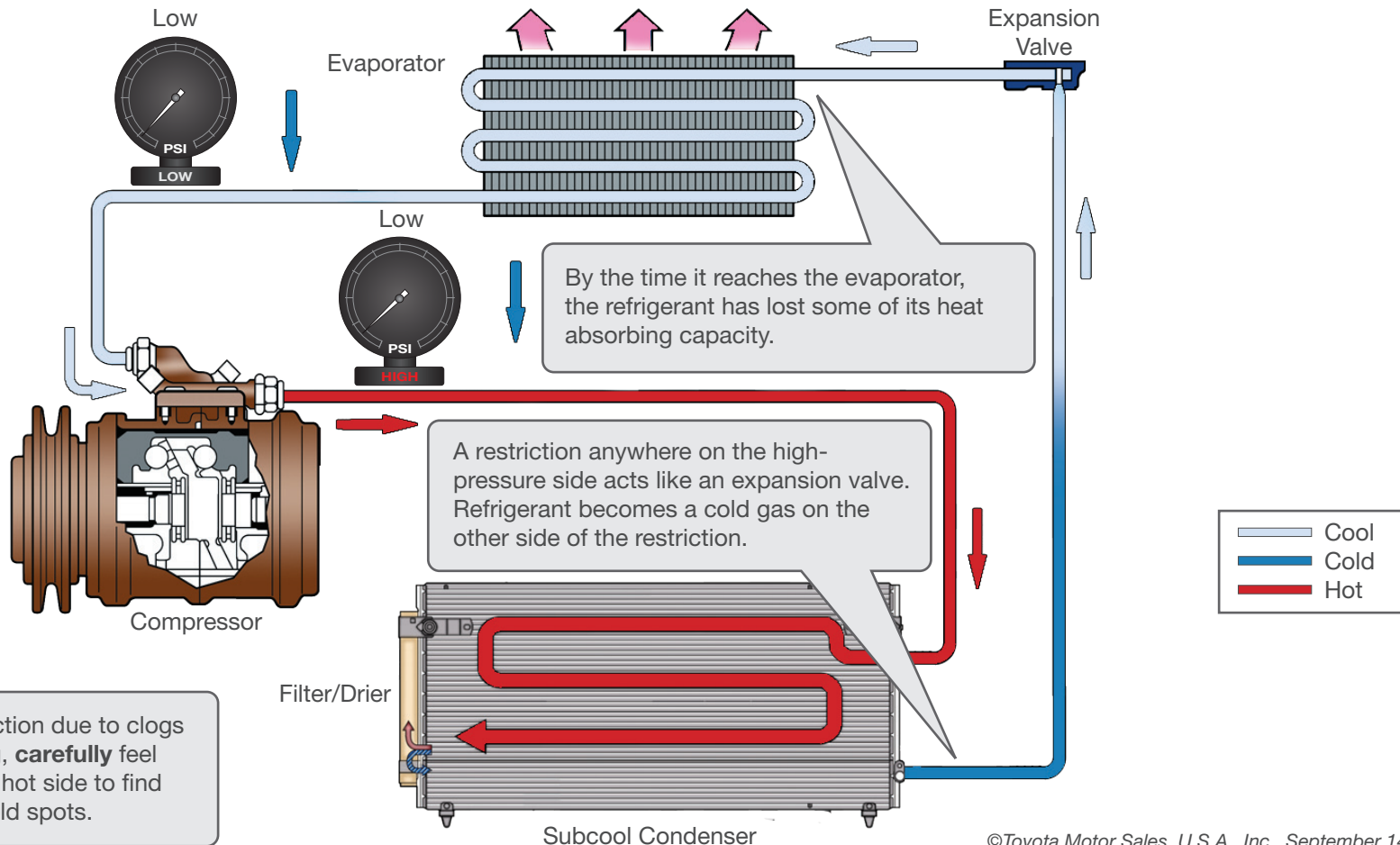
Moisture in System

Restriction in Condenser

Restriction in Condenser

Symptoms:

- Poor cooling
- Low side **below** normal
- High side **below** normal
- Frost or dew on tubing from condenser



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